ISC COMPUTER SCIENCE – TEST II (TOTAL: 60 MARKS)

SECTION A – MCQs (20 × 1 = 20 marks)

1. Which class in Java is most commonly used for reading text files?
   a) FileInputStream b) FileReader c) Scanner d) DataInputStream
2. The extension for a Java package directory corresponds to:
   a) .pkg b) folder name c) .javapkg d) .class
3. Which logic gate outputs **1 only when all inputs are 1**?
   a) OR b) XOR c) NAND d) AND
4. A recursive function must have:
   a) Two parameters b) A loop c) Base case d) Class name
5. Which stream is best used for binary file writing?
   a) FileWriter b) FileOutputStream c) PrintWriter d) BufferedReader
6. java.util is an example of a:
   a) Class b) Package c) Method d) Interface
7. The output of OR gate is 0 when:
   a) Any input is 1 b) All inputs are 0 c) Inputs differ d) Inputs are equal
8. Recursion uses which data structure internally?
   a) Stack b) Queue c) Array d) Linked List
9. Which method is used to read a line from a text file using BufferedReader?
   a) readWord() b) read() c) readLine() d) next()
10. In packages, the package keyword must be placed:
    a) At end of file b) Anywhere in file c) As first statement d) After import
11. What is the output of XOR gate for inputs (1, 1)?
    a) 1 b) 0 c) Undefined d) Same as OR
12. A recursive function that never reaches a base case will cause:
    a) Compilation error b) Runtime exception c) Infinite recursion d) Logical NOT
13. Which of the following is a one-dimensional data structure?
    a) Matrix b) Queue c) Array d) Tree
14. Which class is required for writing primitive data to binary files?
    a) PrintWriter b) DataOutputStream c) ObjectWriter d) BufferedWriter
15. The NOT gate converts 0 into:
    a) 0 b) 1 c) XOR d) None
16. String objects in Java are:
    a) Mutable b) Immutable c) Temporary d) Dynamic
17. What does import java.util.*; do?
    a) Imports Java library b) Imports all classes in util package
    c) Imports only Scanner d) Imports JVM
18. In Java, arrays are:
    a) Static sized b) Dynamic c) Only characters d) Only integers
19. The truth table with inputs 0,0 → output 1 belongs to:
    a) AND b) OR c) NAND d) NOT

20. Recursion is mainly useful for problems involving:
    a) Hardware b) Repetition c) File handling d) Parallel tasks

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SECTION B – Assertion–Reasoning (5 × 2 = 10 marks)

(Choose: a) Both A and R are true, R explains A
b) Both true, R does not explain A
c) A true, R false
d) A false, R true)

21. **Assertion (A):** A package helps avoid name conflicts.
    Reason (R): Packages provide namespaces for organizing classes.
22. **Assertion (A):** Binary files store data in human-readable form.
    Reason (R): Binary files use ASCII representations of characters.
23. **Assertion (A):** All recursive algorithms can be written iteratively.
    Reason (R): Recursion and iteration both rely on repetition.
24. **Assertion (A):** NAND gate is a universal gate.
    Reason (R): Any logic gate can be formed using only NAND gates.
25. **Assertion (A):** Strings in Java cannot be modified once created.
    Reason (R): String class in Java is immutable.

SECTION C – One-sentence Answers (5 × 2 = 10 marks)

26. What is a base case in recursion?
27. Define a package in Java.
28. State one difference between binary and text files.
29. What is the purpose of a NOT gate?
30. What is an array index?

SECTION D – Programming Questions (4 × 5 = 20 marks)

(Attempt any four. Each carries 5 marks.)

31. **Write a Java program to read a text file and count the number of lines present.**
32. **Write a recursive function to compute the sum of digits of a number.**
33. **Write a Java program to input 10 integers into an array and output the smallest and largest values.**
34. **Write a Java program using packages:**
    - **Create a package shapes,**
    - **Inside it create a class Area with a method to compute area of a circle,**
    - **Access it from another class.**
35. **Write a Java program to count the number of vowels in a given string.**